



Including people in the automatic text adaptation process: empirical findings from students with intellectual disability

Åsa Elwér, Evelina Rennes, Karin Nilsson, Lina Cederborg, Daniel Holmer & Arne Jönsson

To cite this article: Åsa Elwér, Evelina Rennes, Karin Nilsson, Lina Cederborg, Daniel Holmer & Arne Jönsson (01 Aug 2025): Including people in the automatic text adaptation process: empirical findings from students with intellectual disability, *Disability and Rehabilitation: Assistive Technology*, DOI: [10.1080/17483107.2025.2536701](https://doi.org/10.1080/17483107.2025.2536701)

To link to this article: <https://doi.org/10.1080/17483107.2025.2536701>



© 2025 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group



[View supplementary material](#)



Published online: 01 Aug 2025.



[Submit your article to this journal](#)



Article views: 186



[View related articles](#)



[View Crossmark data](#)

RESEARCH ARTICLE

 OPEN ACCESS



Including people in the automatic text adaptation process: empirical findings from students with intellectual disability

Åsa Elwér^a, Evelina Rennes^b, Karin Nilsson^a, Lina Cederborg^a, Daniel Holmer^b and Arne Jönsson^b

^aDepartment of Behavioral Sciences and Learning, Linköping University, Linköping, Sweden; ^bDepartment of Computer and Information Science, Linköping University, Linköping, Sweden

ABSTRACT

Modern automatic text adaptation often fails to meet the specific needs of targeted readers. The TextAD project aimed to address this gap by creating tailored texts for students with intellectual disability (ID), focusing on their unique challenges. This exploratory study presents findings from three rounds of testing automatic adaptations at lexical, syntactic, and discourse levels in expository texts. Reading comprehension, self-reported ratings of comprehension, interest, and perceived difficulty, along with metacognitive insights were assessed with 27 students with ID, refining adaptations based on each iteration's results. While comprehension gains for the adapted versions were not significant overall, promising outcomes appeared with lexical and syntactic adaptations in the later rounds of testing, although group heterogeneity influenced results. Despite reporting high comprehension, students noted persistent difficulty understanding many expository concepts. Alongside evaluating the effects of text adaptations on comprehension and perceived understanding, this paper offers insights into assessment approaches for developing automatic adaptations tailored to this audience.

ARTICLE HISTORY

Received 1 November 2024
Revised 23 May 2025
Accepted 12 July 2025

KEYWORDS

ATA; automatic text adaptation; intellectual disability; lexical adaptation; reading comprehension; secondary school; syntactic adaptation

► IMPLICATIONS FOR REHABILITATION

- Automatic lexical and syntactic text adaptations can help improve reading comprehension for students with intellectual disabilities (ID), but significant group heterogeneity affects outcomes.
- It was unclear if automatic discourse-level adaptations improved the student's unsupervised reading, such adaptations may require explicit instruction.
- Students with ID report considerable difficulty understanding concepts in expository texts, suggesting that key vocabulary should be discussed or introduced closely alongside the reading task. Definitions provided in or outside the text did not aid student understanding.

Individuals with intellectual disability (ID) often face significant barriers to participation in society, including challenges in accessing written information. These difficulties can contribute to broader exclusion, particularly in the transition from school to adulthood. In Sweden, 24% of students graduating from upper secondary school for students with ID were classified as Not in Employment, Education, or Daily activities (NEED) [1]. A follow-up study highlighted that problems navigating formal support systems played a central role in these outcomes [2]. Making written information more accessible is, therefore, an important step toward greater inclusion. This study investigated the potential of Automatic Text Adaptation (ATA) to generate Easy Language texts tailored to the needs of secondary school students with ID.

Reading ability in individuals with intellectual disability

ID is defined by general cognitive impairments, with standardised scores below 70 on IQ tests, in combination with difficulties with social and adaptive abilities [3]. Persons with ID exhibit delays in several

CONTACT Åsa Elwér  asa.elwer@liu.se 

 Supplemental data for this article can be accessed online at <https://doi.org/10.1080/17483107.2025.2536701>.

© 2025 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group

This is an Open Access article distributed under the terms of the Creative Commons Attribution License (<http://creativecommons.org/licenses/by/4.0/>), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited. The terms on which this article has been published allow the posting of the Accepted Manuscript in a repository by the author(s) or with their consent.

aspects of cognition and language [4–6], and in the development of reading skills such as decoding [7] and reading comprehension [8]. Students with ID develop decoding skills more slowly and typically reach levels comparable to much younger peers [7]. Development often plateaus early, and decoding remains non-automatized for many individuals with ID [9]. Limitations in decoding ability and language skills, such as vocabulary, cause difficulties with reading comprehension [5]. Individuals with ID particularly struggle with inferential comprehension [10,11]. This type of comprehension requires constructing meaning beyond the literal text by integrating prior knowledge and contextual cues to form a coherent mental model [12]. Metacognition refers to the ability to monitor and regulate one's cognitive processes. This ability is often impaired in individuals with ID [13,14], further affecting their reading comprehension.

Even though decoding and reading comprehension are areas of difficulty for persons with ID, they can improve their literacy skills with suitable materials and interventions [15–18]. However, it should not be expected that persons with ID can benefit from reading instruction to the same extent as typically developing children. Therefore, it is essential to develop strategies that enhance text accessibility for individuals with ID to bridge the gap between individual abilities and societal requirements.

Easy language

Easy Language guidelines aim to make written content accessible to people with cognitive disabilities or limited reading skills. Crossley et al. [19] highlighted the importance of distinguishing between two levels of readability: *micro-level readability*, which concerns the clarity of individual words and sentences, and *macro-level readability*, which involves the coherence and overall structure of the text. Easy Language guidelines address both of these levels, but tension can arise between them—for example, simplifying sentences at the micro level can inadvertently reduce coherence at the macro level [19].

Easy Language guidelines operate across: word, sentence, and text/discourse levels [20–23]. At the word level, the use of common, familiar vocabulary is recommended, while abstract or complex terms should be avoided. Sentence-level adaptations focus on syntactic simplification such as making sentences shorter, more direct, and clearer. This includes discouraging the use of passive constructions and nominalizations to ensure it is always evident who is doing what. Adaptations at the word and sentence level influence the micro-level readability of the text.

Macro-level readability is influenced by adaptation at the discourse level. Easy language guidelines emphasize logical organization of content and clear presentation of key information. This may include strategies such as placing important content early, using headings, and visually highlighting critical points. Easy Language guidelines are commonly used in many contexts; however, as pointed out by Wengelin [24], the empirical grounding of such guidelines is scarce. Moreover, the guidelines often describe Easy Language as a single feature, disregarding the heterogeneity of the reader groups that may benefit from Easy Language text.

Automatic Text Adaptation (ATA) refers to the use of computer programs often using AI to automatically simplify texts, making them more accessible and readable. The ATA systems typically rely on rule-based methods or statistical models trained on examples of simplified texts. In contrast, manual text simplification is carried out by humans who follow general guidelines but also draw on their own judgment, taking into account factors such as the reader's background, purpose, and context. While manual simplification tends to be more flexible and context-sensitive, ATA offers the advantages of lower cost and scalability, making it possible to apply adaptations to virtually any text [25].

Techniques that can be used for this purpose are word simplification, sentence simplification, and summarization, but they also include other techniques such as identifying and explaining keywords, spelling out abbreviations, or changing the typography of the text to make it clearer and more accessible to the reader. ATA is today primarily data driven which means that current approaches to text adaptation rely mainly on large collections of real-world text data rather than manually crafted rules. To evaluate the quality and characteristics of automatic adaptations, a set of standard metrics is commonly used: BLEU [26], SARI [27], and SAMSA [28]. These measures assess quality in terms of semantic similarity between the adapted and the original text, grammatical correctness, and the degree of textual modification. However, these measures do not account for whether the adaptations are easier to read for the

intended target group. This highlights a clear research gap, as few studies have investigated whether ATA are genuinely helpful to the readers they are intended to support [25].

Text adaptations for individuals with intellectual disability

A few studies have explored the effect of Easy Language materials on persons with ID. Fajardo et al. [10] found that participants with ID gave a high percentage of correct answers to comprehension questions of Easy Language news articles. Similarly, Karreman et al. [29] found that language adaptations of a website, following Easy Language guidelines, proved useful to improve comprehension for adults with ID compared to an original version. Säuberli et al. [30] found that comprehension of manually adapted texts was better compared to original texts in participants with ID. However, automatically adapted texts did not help the participants. The authors suggested that this pattern may be due to errors in the ATA versions and a better alignment between the questions and the original or manually adapted texts compared to the automatically adapted ones.

However, studies have also failed to find effects of text adaptations for individuals with ID. Adding connectives and substituting rare words with more frequent ones [31], adding symbols [32], or pictures [33] are examples of unsuccessful adaptations. One small-scale Swedish study found that some Easy Language texts provided by public authorities were difficult to read for persons with ID [34], as the content was too abstract, given the capacity of the readers. The same pattern of results on Easy Language versions of authority texts was found in a study by Frondén and Kaakinen [35]. Several researchers have questioned whether conventional guidelines for Easy Language are sufficient for participants with ID, indicating a need for further investigation into this area [10,24].

Present study

This exploratory study aimed to evaluate how students with ID read and understood ATA texts compared to the original versions. We used an iterative approach, including several assessments, to gain as much information as possible about the benefits and challenges of ATA texts. The study was carried out in three student groups, recruiting nine participants for each iteration. The idea was to improve the adaptations from one iteration to the next, based on previous results. The following research questions were addressed: 1) To what extent do different methods of ATA aid comprehension in students with ID? 2) How do students evaluate the original and adapted texts with regard to interest, perceived reading comprehension, and level of difficulty? 3) How do students describe their experiences and challenges after having read the ATA and original texts? 4) How can different types of assessments inform future text adaptations for students with ID?

Method

Participants

In Sweden, students with an ID diagnosis are allowed to attend a compulsory school for students with ID. We contacted five schools and teachers working with student groups with mild to moderate intellectual disability who were deemed able to engage with adapted versions of the national curriculum. Teachers were asked to distribute consent forms to students in grades 7–9 who they considered capable of reading simple paragraphs. Each iteration included nine students, with a total of 27 students participating in the study (11 girls in total, with 1, 5, and 5 girls in each respective iteration). The participants were between 13 and 17 years old. The mean ages for the three iterations were 15.47 ($SD=0.96$), 14.47 ($SD=1.32$), and 15.03 ($SD=0.59$), respectively. The mean value for iteration 3 is based on only five participants due to missing data. The project was approved by the Swedish Ethical Review Authority (Dnr: 2021-02918).

The cognitive and language skills of the three groups of students included in separate iterations in the study are presented in Table 1. The mean decoding ability of the participants corresponded to a

Table 1. Means, standard deviations and ranges of cognitive and language tests in all student groups taking part in the three iterations.

	Student group 1		Student group 2		Student group 3	
	<i>M(SD)</i>	Range	<i>M(SD)</i>	Range	<i>M(SD)</i>	Range
Vocabulary ^a	122.44 (27.49)	86–159	114.44 (18.76)	83–138	118.44 (25.48)	83–161
Decoding ^b	119.00 (24.78)	83–168	116.00 (31.15)	55–155	115.78 (29.67)	60–157
Grammar ^c	10.77 (3.49)	4–15	11.33 (6.18)	1–17	13.00 (3.30)	6–16
Digit Span ^d	6.22 (1.30)	4–8	7.44 (1.94)	5–10	5.89 (1.61)	4–8

Note. A=Peabody Picture Vocabulary test [36]; b=LäSt [37]; c=Test of the reception of grammar [TROG] [38]; d=Wisc Digit span [39].

reading age of 10 years and 3 months for group 1 and 10 years for group 2 and 3. For grammatical understanding, the age equivalent for the mean value is 6 years and 2 months for groups 1 and 2, and 6 years and 11 months for group 3. For the test of working memory, the digit span, the mean results were below the average range, which is 8–12. There are no official norms for the TROG in Swedish. As can be seen in Table 1, the participants differ quite a lot in their performance.

Reading comprehension tests

Four texts with five questions each were used to examine reading comprehension of original and adapted texts in the target group. The texts were taken from age-appropriate domestic science and geography textbooks and covered different subject areas, but the content was concrete and considered to be familiar to the students (food, health, and countries). Specifically, the texts were about legumes, exercise [40], Brazil, and Australia [41]. The texts were between 200 and 250 words. Original texts translated into English are included in Supplementary Appendix 1.

The texts were presented using formatting principles suggested by the Swedish Agency for Accessible Media [23] to make them accessible to poor readers. Garamond font size 14 was used with 3-point spacing between lines, blank rows between paragraphs, wide margins, and no justification of text. In addition, all pictures were removed as the focus of this study was on the text. Images can give away meaning or answers, shifting focus from reading to interpreting visuals. This reduces validity and may increase cognitive load, especially for students with cognitive impairments. Headings and subheadings, when included, were kept and remained consistent across the different text versions.

Five multiple-choice questions were created for each text. Two of them required making inferences, while the remaining three addressed explicitly stated information. Each question had four response alternatives. The same questions were used for all three test iterations, with just minor adjustments in alternatives, to reflect the specific text adaptations. For instance, if a word was replaced by a synonym in one version of the text, the corresponding question used that same synonym. This ensured that all students answered questions that were linguistically aligned with the version of the text they had read. The questions for the original versions of the texts are included in Supplementary Appendix 1.

Structured oral questionnaire

A structured oral questionnaire was completed after each text reading to gather information about the participants' post-reading perceptions of the texts and metacognitive insights. The questionnaire consisted of eight structured questions (Supplementary Appendix 2). The first three questions examined post-reading perceptions of texts and used a scale of 1–5. The students rated the text's perceived difficulty (1=very easy to 5=very difficult), their self-assessed level of comprehension (1=nothing to 5=everything), and their interest in the text (1=very boring to 5=very interesting). Students were asked to elaborate on their choice of rating and provide reasoning behind their choices after each of these questions. The following five open questions focused on metacognitive aspects of the text, i.e., aspects that they found challenging, such as words and sentences that were difficult to decode or understand. For these questions, the students were encouraged to revisit the text to refresh their memory of the challenging parts and to explain the reasoning behind their answers.

Implemented adaptations

In the first iteration of the study, we used general Easy Language adaptations which are common across guidelines [20–23]. Descriptions of the text versions and the adaptations are included in Supplementary Appendices 3 and 4.

Lexical simplification

The lexical simplification module aimed to find and replace complex words. For the first iteration, words that did not have an entry in the Swedish dictionary SWEVOC [42], indicating that the word is potentially outside the base vocabulary for a Swedish speaker, were selected as candidates for synonym replacement. If this word had a synonymous equivalent in the Swedish synonym lexicon SYNLEX [43] that was more frequently occurring (according to the additional frequency annotations to SYNLEX performed by Keskisärkkä and Jönsson [44], the word was replaced with the synonym.

Syntactic simplification

We used a set of syntactic simplification rules working at the sentence level. The rule set implemented in the first iteration is presented in Table 2 and has been previously used in various Swedish text simplification projects [45,46].

Procedure

One member of the research team met with each of the students individually on three occasions. During the first two sessions, reading comprehension of original and adapted texts was examined. All participants read two original and two adapted texts each and answered 20 comprehension questions (five per text). During the first test session, each student read two texts within the same subject area (e.g., domestic science), one in its original version and one adapted. The same procedure was followed in the second test session, but using texts from a different subject area (e.g., geography). The order and version of the texts were counterbalanced across participants. According to their preference, the students read the text aloud or silently, and they answered the reading comprehension questions independently. They were allowed to go back and look at the text while answering the questions. Following the reading of each text, the oral questionnaire was administered. The last test session focused on testing cognitive abilities and language skills.

Analysis

The results for each participant were averaged across the two original versions and the two adapted versions for each subject. Text versions were compared using paired t-tests and Cohen's *d* to estimate effect size. The same procedure was used for the ratings regarding the post-reading perception of the texts (ratings between 1 and 5 on the first three questions of the questionnaire). With a small sample

Table 2. The initial set of syntactic adaptations applied in the first iteration.

Rule	Definition
Proximisation	Changes the text so that it is psychologically closer to the reader by addressing the reader directly when possible. <i>Example: One should always try to stay calm in difficult situations. ->You should always try to stay calm in difficult situations.</i>
Passive-to-active	Rewrites a sentence in passive tense to active tense. <i>Example: The meeting was attended by all the members. -> All members attended the meeting.</i>
Quotation inversion	Makes sure that an utterance is situated after the person saying in the text. <i>Example: "You should study more for the exam," the teacher said.->The teacher said, "You should study more for the exam".</i>
Straight word order	Rearranges the words in a sentence to a subject-verb-object word order. <i>Example: Only after reviewing all the data did the team identify the underlying trend. -> The team identified the underlying trend only after reviewing all the data.</i>
Sentence split	Divides long and complex sentences. The split rules split the sentence at conjunctions, relative clauses, and appositions. <i>Example: The teacher explained the lesson clearly, but some students still didn't understand, so she offered extra help after class.->The teacher explained the lesson clearly. Some students still didn't understand. She offered extra help after class.</i>

size in each iteration, significance can only be expected if differences are large, which was not to be expected in this case based on previous results. Therefore, we interpreted small and moderate effect sizes as potentially valid differences between text versions.

The students' verbal responses to the open-ended questions in the questionnaires, including their reasoning behind their choices for questions 1–3 and their answers to the metacognitive insight questions (4–9), were reviewed and systematically analysed. Recurring patterns and notable observations were identified and organized into thematic categories. These themes were used to address the research question: How do students describe their experiences and challenges after having read the ATA and original texts? The student responses were analysed to interpret their understanding of the statements.

Results iteration 1

This section presents the results of the first test iteration of the project. First, we present results from the reading comprehension tests. Second, we present results from the oral questionnaires under two separate headings, one for the post-reading perception of text, which includes the ratings (1–5) and one for the metacognitive insights which included verbal explanations.

Reading comprehension

The nine participants' reading comprehension scores were similar across the two text versions. The comparisons revealed a negligible effect size and favoured the original version $t(8) = 0.45$, $p = 0.67$, $d = 0.15$. The mean results, averaged across two texts with a maximum score of 5, were $M = 2.94$ ($SD = 0.92$), 59% correct for the original and $M = 2.78$ ($SD = 0.62$), 56% correct for the adapted versions. The average scores, below 3 points for both versions, suggested relatively limited reading comprehension; the means ranged from 1.5 to 4 across participants and highlighted significant variability in the results.

Post-reading perceptions of text

The average perceived difficulty ratings ($M = 2.39$, $SD = 0.99$ for the original texts and $M = 2.06$, $SD = 0.68$ for the adapted) indicated that students found both versions moderately easy, and the adapted texts were slightly easier. The effect size was small, $t(8) = 1.11$, $p = 0.30$, $d = 0.37$. For self-assessed level of comprehension, ratings were relatively high, with mean scores near 4 for both versions ($M = 3.78$, $SD = 0.80$ for the original; $M = 4.17$, $SD = 0.66$ for the adapted). The comparison showed a moderate effect size between versions, $t(8) = 1.98$, $p = 0.09$, $d = 0.65$, and suggested better self-assessed comprehension of the adapted texts. Lastly, students' ratings of their interest in the texts were similarly high ($M = 4.28$, $SD = 0.94$ for the original; $M = 4.33$, $SD = 1.12$ for the adapted), with a negligible effect size, $t(8) = 0.22$, $p = 0.83$, $d = 0.07$, which indicated no differences in perceived interest between versions. In summary, the students found the texts relatively easy to read, felt they understood most of the content, and considered the texts interesting, regardless of the version. Potential advantages of the adapted texts were observed in the ratings of difficulty and self-assessed comprehension. These advantages did not correspond with the reading comprehension tests.

Metacognitive insights

Most of the students' comments were relevant to both versions of the text (the original and the adapted version). In cases where the students commented on a feature present only in one version, we have stated which version the student talked about. Aspects that influenced the reading experience are presented under the headings *Positive perceptions of the texts*, *Limited content in sentences makes it easier to read*, and *Difficult words as roadblocks*. In the results, Swedish words and sentences have been translated into English. When necessary, specific words or expressions are presented in Swedish, followed by their English translation in parentheses.

Positive perceptions of the texts

Overall, the students were very positive about the texts, describing them as interesting and easy to follow. They were excited about learning new ideas and praised the clear structure of the texts. During interviews, they also offered suggestions to enhance comprehensibility.

The students remarked that having an interest in the subject made the text easier to understand. One student specifically mentioned that the text on exercise was easier compared to others, such as the one on legumes, because they were more interested in exercise. The student explained, “None were difficult, but the one about exercise was easier because I’m interested in it.” This suggests a connection between personal interest, motivation, and how challenging a text is perceived to be.

Several students wanted to share something they had learned from the text. The student in the quote below experienced the reading as enjoyable, as the content gave the student more knowledge. “It was fun to read, I like facts. Now I know almost everything about Australia. I know that they celebrate Christmas on the beach.” In the text, the author presents many differences between Sweden and Australia and states that their warmest period corresponds to the coldest period in Sweden. Hence, you can spend Christmas at the beach. The student described that they had learned a lot and felt that they knew almost everything about Australia. The quote indicates that the student felt a sense of satisfaction at having learned something new.

Some students highlighted structural aspects when describing texts and why they liked them. They mentioned headings and sub-headings, but also that it was positive that the text dealt with one thing at a time. One student stated: “Good with headings and sub-headings so you don’t mix everything up. They talk about one thing at a time”. The student pointed to both the importance of structure and clarity in the layout of the text. Headings and sub-headings made the text easier to read and facilitated finding information. Adhering to a single topic at a time results in a more comprehensible and organized text.

The students were not only positive regarding the texts. One student suggested marking keywords. “You can write in black but have certain words that are pink or some other color so it will be easier for those who have difficulty seeing and for everyone, so it will be easier to see where you are in the text”. The student expressed a wish for tools that make reading easier and suggested that some words could be marked with a different color. This would both help students who need it because of visual impairment and help all readers to better find information in the text. This suggests that the student recognized a need to scaffold the reader in identifying the key points of the text.

Another student’s suggestion was to include pictures to illustrate certain aspects of the text. One student said: “I would like a picture that shows what different vitamins are like vitamin B and what differentiates different vitamins.” Some texts in the textbooks originally included pictures, but these were removed for this project to emphasize isolated reading. The student’s statement could be seen as an expression of the need to have more background knowledge. They appeared to want more information about vitamins, what they are, and how they differ from each other. The text does not explain what vitamins are.

Limited content in sentences makes it easier to read

When the students were asked to go back through the text retrospectively to find and give examples, most of the students responded that they did not find any sentences difficult. Only two aspects at the sentence level were commented on by students: sentence length and the use of commas. Short sentences were preferred by several students. After having read the original version of *Legumes*, one student expressed that “the text about legumes should have more periods”. Another student found commas disturbing when reading, especially if there were several commas in the same sentence. When the student read a sentence with six types of legumes listed, she experienced that her reading was disrupted. The student explained: “There are too many commas. I’d rather have *for example*, and two examples and not commas. The sentence gets a little messy. There were too many examples”. One student expressed this as: “In my head, it was too much, when I read everything got mixed up. I needed to digest the text and think”. This student’s statement, the dislike of commas and preference for short sentences, may be an expression of the same phenomenon, namely cognitive overload in trying to process a lot of information at the same time.

Difficult words as roadblocks

Word difficulties were commonly highlighted when students explained how easy or difficult the texts were. The link between word comprehension and perceived difficulty appears to be strong. In the questionnaire, the students were asked to indicate words that were difficult to read and words that were difficult to understand. For many students, the distinction was unclear, and they often referred to the same words when describing both aspects.

Some students talked about the importance of understanding the context and that the understanding of an entire text is lower if the text contains difficult words. It seems that the difficult words can cause the reading flow to stop entirely. “I just focused on the fact that I didn’t know the word ‘hemisphere’. Then the whole sentence became difficult, and I had to read it again”. The student described that getting caught up on individual words can disrupt the entire reading process. It appears that it was difficult for the student to shift focus and work around the difficult words, as the word became a disturbing element.

The students highlighted many words that they had not heard before and did not know the meaning of. These words were often subject-specific, and closely linked to what the text was about. The students also expressed that some of the texts dealt with content that they did not know anything about. As an answer to the question if any words were difficult to read, one student stated: “*Sparse, sparser, sparsely populated, eucalyptus trees* – I had never heard those words”. Some other examples from other texts were *kidney beans, protein source, endorphins, obesity, ungulates, hemisphere, suburb, and inhabitant*. As these words are subject-specific, they are difficult to replace, as there may not be very many true synonyms. However, *sparse* was substituted with *more scattered* in the adapted version. The understanding of the concepts is closely tied to the subject knowledge of the texts. It appears that the context of the texts was not clear in all cases, which makes the texts much more demanding for any reader.

The students also gave examples of reasoning about the meaning of unknown words. This quote came after the student had read the original text *Australia*. “I understood the introduction, but then I got to *glest befolkat* (Eng: sparsely populated) and it was difficult. *Glest* (Eng: sparsely) should mean *flest* (Eng: the most)”. In the original text, the student encountered a sentence where Australia was compared to Sweden, as they are both sparsely populated. The student realized that *sparsely* was an unknown word and tried to figure out the meaning. Here, they suggested a similarly spelt word that is very different in meaning. For a typical reader, the context described with deserts and dry grasslands, which makes it impossible to live in those areas, would be clues to figure out the meaning of the word *sparse*. However, that information did not help the student. To figure out the meaning of a word by connecting it to a similarly spelt word is an immature reading comprehension strategy, which suggests a focus on the word level rather than the text level.

In Swedish, compound words can be quite long, as they can consist of combinations of long words, or three or more words written together. Some examples from the used texts were *kontorsskyskrapor, lyxbostäder, tjocktamscancer* and *eukalyptusträd* (Eng: office skyscrapers, luxury homes, colon cancer, and eucalyptus trees). One student explained that they read too fast and could not decode the words at that speed. Adapting reading speed to the difficulty of the words, does not seem to be done automatically for the student. One student commented on the word *tjocktarmscancer* (Eng: colon cancer), which starts with the /ç/ sound in Swedish: “I know the /ç/ sound, but when the words are long, I lose my way, and I have to read again”. The /ç/ sound is a voiceless alveolo-palatal fricative which is an exception to Swedish spelling rules. The sound can be spelt in several different ways that you must learn by heart. The student in the example above is familiar with the sound and has most likely practiced the connection between the spelling and the sound. The student’s response indicated that decoding still hindered reading comprehension, as the focus remained on spelling conventions during the discussion of the text. To sum up, at the word level, the students experienced difficulties with unknown specific words, long words, and irregularly spelled words.

Methods iteration 2

In iteration 2, we retained the same texts but introduced new adaptations and added a reading comprehension task using the Sentence Verification Technique (SVT) [47]. This was because low scores on the multiple-choice test raised concerns about its reliability for our students. The multiple-choice format,

requiring comparison of four answers, may have been too complex given their limited attention and working memory, potentially leading to guessing. To address this, we included a simpler assessment.

The SVT measures literal comprehension through a straightforward yes/no question: “Do you recognize this content from the text?”. Four types of sentences are presented: (a) original sentences from the text, (b) meaning-altered sentences where the meaning is changed (e.g., “Monaco is one of the biggest countries” instead of “smallest”), (c) distractor sentences on the same topic but not in the text, and (d) paraphrased sentences with the same meaning as an original sentence from the text. The task focuses on understanding the surface meaning of the texts [48]. For our participants, we simplified the format by including four original sentences, two meaning-altered sentences, and two distractors, eight questions per text. This allowed for easier instructions and simply asking “Do you recognize this sentence from the text?” without distinguishing paraphrases. The test items (sentences) were matched to the text version each student had read.

The results from iteration 1 indicated that the initial set of adaptations was insufficient, as they did not enhance students’ reading comprehension of the adapted texts, and word-level difficulties appeared to hinder comprehension. If comprehension is hindered by word-level difficulties, additional adaptations are unlikely to yield substantial improvement. For these reasons, we focused on the word level in the second iteration.

Implemented adaptations iteration 2

First, we replaced SWEVOC with an extended version of NYLLEX [49]. The rationale behind this change was to be able to exclude more basic words not necessary for replacement, since NYLLEX contains almost twice the number of entries compared to SWEVOC. Consequently, we could be less restrictive in the following steps of the replacement process, since the probability that the word considered for a replacement was a word outside of the base vocabulary was higher.

Second, we changed the approach to how to select a synonym in SYNLEX. In iteration 1, we selected the possible synonym with the highest frequency. In iteration 2 we implemented a Swedish Fasttext [50] word vector model to select the synonym with the highest semantic similarity, from all of the entries in SYNLEX with a higher frequency than the word currently considered for replacement. This also allowed us to filter possible synonyms that semantically were too far from the original word according to the Fasttext model. Taken together, these two changes made it possible for the automatic adaptation tool to select from a wider range of candidate synonyms, while also providing additional filtering steps to ensure a higher quality of the suggested synonym.

Epithets, such as *the country Australia* (rather than Australia), were added to the text [51]. The epithet module was able to identify geographical epithets, organizations, and persons, but in the final texts, only the geographical epithets were identified and added to the adapted texts. This change was motivated by the fact that some words were still too difficult to understand for the students with ID. To use explanatory epithets is an advice given by the Swedish Agency for Accessible Media [23].

We changed the syntactic simplification framework to use the simplification framework developed by Priscepov [52]. We made two modifications to the automatic adaptation tools to better align with our objectives. First, we refined the syntactic simplification rules to minimize errors. Second, we deactivated one rule, the Time Adverbial, which occasionally introduced errors into the texts. Descriptions of the texts and adaptations are included in Supplementary [Appendices 3 and 4](#).

Results iteration 2

Following adaptation improvements, original and adapted texts were tested on a new group of nine students, assessing reading comprehension (using two tests), post-reading perceptions, and metacognitive insights.

Reading comprehension

Comprehension of the adapted texts ($M=2.33$, $SD=0.90$, 46% correct) was somewhat higher compared to the original texts ($M=1.94$, $SD=0.88$, 39% correct) on the multiple-choice items. The difference did

not reach significance but suggested a small effect size ($t(8) = 1.0$, $p=0.35$, $d=0.33$). This suggests that, unlike in iteration 1, the adaptations appeared to be helpful.

The same pattern was found for the SVT tests ($M=6.06$, $SD=0.68$, 76% correct for the original and $M=6.28$, $SD=0.83$, 79% correct for the adapted versions). The differences between versions again favoured the adapted texts with a small effect size ($t(8) = 0.66$, $p=0.53$, $d=0.22$).

Post-reading perceptions of texts

Just as in iteration 1, the students rated their perceptions of the text. The students generally considered the texts relatively easy to read, but this time they rated the original versions as easier ($M=1.83$ $SD=0.79$) compared to the adapted texts ($M=2.28$, $SD=0.94$). The effect size was small ($t(8) = 1.06$, $p=0.32$, $d=0.35$). A small effect size was also found between the two versions for self-assessed comprehension ($t(8) = 1.08$, $p=0.32$, $d=0.36$) which showed that the students felt that they understood more of the original version ($M=4.08$, $SD=0.73$) compared to the adapted version ($M=3.86$, $SD=0.70$). In addition, the students rated the original texts as more interesting compared to the adapted versions with a small effect size ($M=4.28$, $SD=0.80$ for the original, $M=4.08$, $SD=0.95$ for the adapted, $t(8) = 0.64$, $p=0.54$, $d=0.21$). Thus, all the post-reading perception ratings favoured the original version, a pattern not reflected in the reading comprehension results.

Metacognitive insights

The answers to the open questions regarding experiences and challenges in the texts in iteration 2 mirrored those from iteration 1, with students mainly focusing on word-level issues. Despite improvements in the synonym function, some complex words (e.g., legumes, endorphins) were difficult to simplify. Sentence- or text-level insights remained rare.

Methods iteration 3

Based on the results from iteration 2, we concluded that we had reached the limits of synonym replacement and sentence-level simplification for these texts. Therefore, in the third iteration, we shifted our focus to alternative strategies for addressing challenging vocabulary and introduced two discourse-level adaptations along with a typographic adaptation to further support comprehension. These changes were motivated by results from previous iterations, which indicated that vocabulary continued to pose a challenge for many readers, as reflected in their responses to the metacognitive insight questions. Also, the overall comprehension, as measured by the multiple-choice questions, remained low. Additionally, student feedback revealed a dislike for sentences with multiple commas.

We added questions to the questionnaire, asking explicitly about the evident features of the adapted texts in iteration 3 (summaries, lists with bullet points, and word explanations). We also asked the students to motivate their answers. Descriptions of the texts and adaptations are included in Supplementary Appendices 3 and 4.

Implemented adaptations

Challenging vocabulary was dealt with in two different ways in iteration 3. First, we improved the epithet function from iteration 2, notably enhancing the number and detail of geographical explaining epithets. For example, we allowed cities in the texts to have specific descriptors. We also added word explanations. In each text, keywords were highlighted and given a short definition on a separate page. The keywords were selected with a system based on KeyBERT [53]. By embedding the text at both document and word levels in a vector space, we could identify individual words with the highest semantic similarity to the text, which should be regarded as keywords. To generate explanations for the keywords, a GPT-SW3 [54] model was used. The model was given a few example pairs of keywords and their corresponding one-sentence explanations, prompting it to generate similar explanations for new keywords.

One discourse-level adaptation introduced in iteration 3 was the inclusion of a summary at the beginning of the text. This was intended to prime students with an overview of the content, helping them

activate relevant background knowledge and better anticipate what the text would be about. The summary consisted of key sentences generated by the extractive summarization model ElsaSum [55], maintaining a length of about 30% of the original text, as suggested by Smith and Jönsson [56]. Highlighting keywords in the text can also be considered a discourse-level adaptation, intended to help students identify the main topics and better understand what the text is about.

Despite the focus on vocabulary and discourse-level adaptations in the third iteration, we included a set of syntactic simplification rules. Specifically, we incorporated a subset of the syntactic adaptations proposed by Decker [57], which involved simplifying noun phrases by removing articles, adjectives, and prepositional attributes. This was an attempt to simplify some of the overly complex sentences.

A typographic adaptation involved identifying sequences of nominal phrases separated by commas, including a concluding 'and/or' before the last phrase. All nominal phrases within these sequences were then converted into separate items presented as bulleted lists within the texts.

Results iteration 3

For the third iteration, a new group of nine students read original and adapted text versions, and the same assessment methods were applied.

Reading comprehension

The mean number of correctly answered multiple-choice questions was higher for the adapted version with a small effect size ($M=1.78$, $SD=1.20$, 36% correct for the original and $M=2.06$, $SD=1.13$, 41% correct for the adapted versions; $t(8) = 0.73$, $p=0.49$, $d=0.24$). The opposite pattern of results was found for the SVT questions ($M=5.28$, $SD=1.06$, 66% correct for the original and $M=5.06$, $SD=1.29$, 63% correct for the adapted versions). The comparisons between versions revealed a negligible effect size ($t(8) = 0.45$, $p=0.67$, $d=0.15$). The reading comprehension results across all three iterations are presented in Table 3. The individual data are presented in Figures 1 and 2, revealing substantial heterogeneity in how the participants responded to the adaptations across iterations.

Post-reading perceptions of texts

The results from iteration 3 showed that ratings were similar for the original and adapted versions for perceived difficulty and the effect size was negligible however in favor of the original versions like in iteration 2 ($M=2.44$, $SD=0.64$, for original; $M=2.58$, $SD=0.64$, for adapted; $t(8)=0.56$, $p=0.59$, $d=0.19$). Just as in iteration 2, the students rated their self-assessed comprehension higher for the original versions, and the effect size was small ($M=3.83$, $SD=0.56$ for original; $M=3.56$, $SD=1.01$ for adapted; $t(8) = 0.89$, $p=0.40$, $d=0.30$). However, they rated the adapted versions as more interesting with a small effect size ($M=3.50$, $SD=1.20$ for original; $M=4.36$, $SD=0.69$ for adapted), $t(8) = 0.88$, $p=0.41$, $d=0.33$). This was the opposite pattern of iteration 2. The post-reading perceptions across all three iterations are presented in Table 4.

Table 3. Reading comprehension results across all three iterations.

Multiple-choice questions						
Iteration	Original		Adapted		<i>t</i>	<i>d</i>
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>		
1	2.94	0.92	2.78	0.62	0.45	0.15
2	1.94	0.88	2.33	0.90	1.0	0.33
3	1.78	1.20	2.06	1.13	0.73	0.24
SVT questions						
Iteration	Original		Adapted		<i>t</i>	<i>d</i>
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>		
2	6.06	0.68	6.28	0.83	0.66	0.22
3	5.28	1.06	5.06	1.29	0.45	0.15

Note. All *p*s were above 0.05.

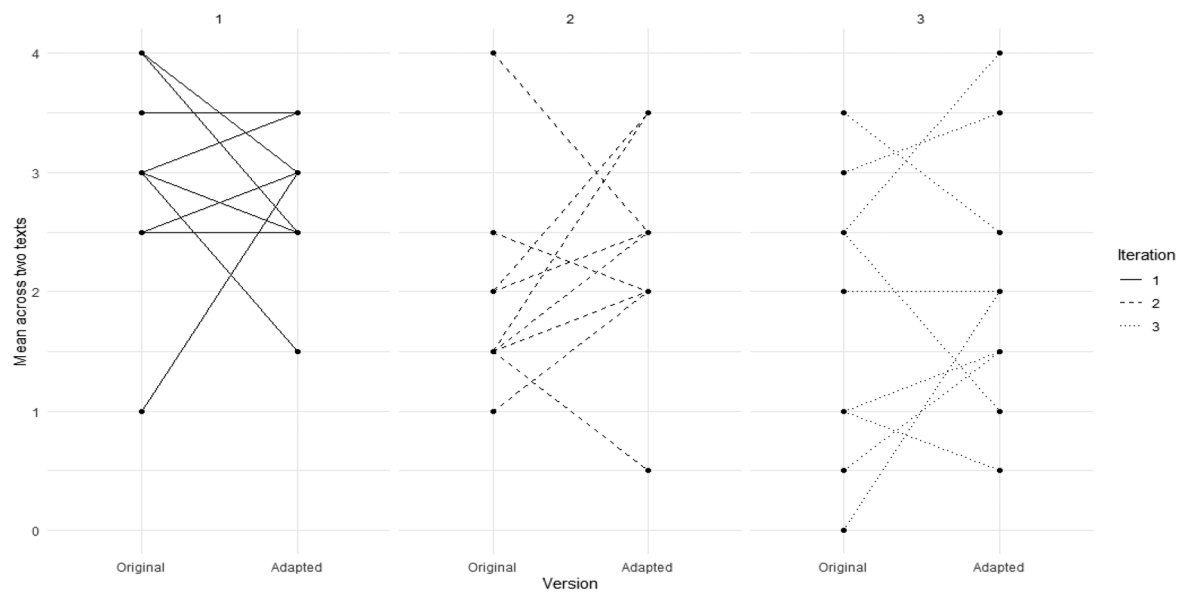


Figure 1. Individual results on multiple-choice reading comprehension tests across iterations.

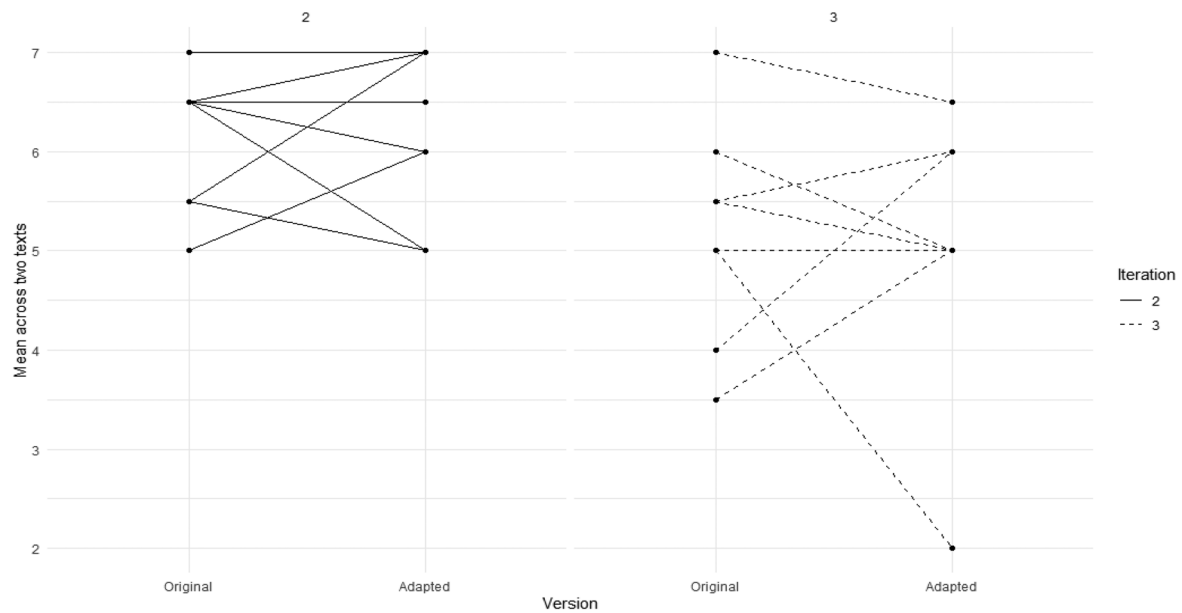


Figure 2. Individual results on SVT reading comprehension tests across iteration 2 and 3.

Table 4. Post-reading perceptions of text for all three iterations.

Iteration		Original		Adapted		<i>t</i>	<i>d</i>
		<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>		
1	Difficulty	2.39	0.99	2.06	0.68	1.11	0.37
	Comprehension	3.78	0.80	4.17	0.66	1.98	0.65
	Interest	4.28	0.94	4.33	1.12	0.22	0.07
2	Difficulty	1.83	0.79	2.28	0.94	1.06	0.35
	Comprehension	4.08	0.73	3.86	0.70	1.08	0.36
	Interest	4.28	0.80	4.08	0.95	0.64	0.21
3	Difficulty	2.44	0.64	2.58	0.64	0.59	0.19
	Comprehension	3.83	0.56	3.56	1.01	0.89	0.30
	Interest	3.50	1.20	4.36	0.69	0.88	0.33

Note. All *p*s were above 0.05.

Metacognitive insights

The differences in the participants' use and perception of evident features of iteration 3 adapted texts (summary, bullet points, and explained keywords) are outlined in the themes: *Positive perception of text-level adaptations*, and *Text-level adaptations are insufficient when reading comprehension is low*.

Positive perception of text-level adaptations

Most participants answered that they thought the adaptations were good when asked a direct question. This included the summaries, the highlighted words with explanations, and the bullet lists. Mostly, the participants did not expand on why these aspects were helpful for them, even when prompted to do so. It is difficult to know if the features made a difference for the participants based on such statements. More specific feedback on the features was given by a few participants. Concerning the summary, one participant stated: "The short version was good, it helped me understand what the text was about". This statement is in line with the idea of including summaries to help the participant activate background knowledge of the text and process it better. The statement does suggest that the participant found it useful in such a way.

Regarding the highlighted and explained keywords, many participants considered this a positive feature, with specific comments noting that "it made the important words in the texts easier to spot". Several individuals revisited the explanations, but one participant mentioned that "it was difficult to remember the explanations." This highlights the challenge of shifting attention from the text to the specific word, reading its definition, and then retaining that definition when refocusing on the text.

Text-level adaptations are insufficient when reading comprehension is low

Some participants' statements indicated that the adaptations were perceived as less helpful than intended. It was noted that the added length of the text, due to the inclusion of the summary at the beginning, made it more exhausting to read. This indicates that reading is laborious and keeping it short may be more important than trying to help comprehension by priming readers to the content.

A few readers mentioned that they overlooked adapted aspects while reading. Also, one participant admitted they did not go back to review the provided explanations, even though they stated that the "black words" were unfamiliar (keywords highlighted in bold). One participant claimed she knew all the words already, including the bolded ones. All these statements suggest that the adaptations were not aligned with the reading process of the participants.

Discussion

In this exploratory study, we used Automatic Text Adaptations (ATA) to modify texts at the word, sentence, and discourse to improve reading comprehension for secondary school students with ID. Below, we address the study's research questions, focusing on: (a) differences in comprehension scores between original and adapted versions of the texts; (b) readers' ratings of interest, perceived reading comprehension and level of difficulty; (c) the challenges readers encountered while engaging with the texts; and (d) how the various assessment methods used in the study can inform future text adaptations for students with ID.

Comprehension of expository text

The initial adaptation employed general Easy Language guidelines as a starting point and modified syntax and vocabulary. The average scores for multiple-choice questions were very similar between both versions of the text, which is in line with previous studies showing that there is no guarantee of adapted texts improving comprehension in readers with ID [31–33]. Although it may seem counterintuitive, there are several reasons why text adaptations might not always be effective. First, adaptations may benefit some readers but not others, as this population is highly heterogeneous. This pattern appears to exist in our study as well, as can be seen in the individual trajectories. One possible explanation is an interaction between reading comprehension level and the effectiveness of adaptations. Fajardo et al. [58] found that

their strongest readers with ID performed better with the original texts. The less proficient readers showed no benefit from the adaptations either, possibly because their reading level was too low to engage meaningfully with the material, adapted or not. In this study, we kept all information from the original texts in the adapted versions but expressed it in simpler terms. Other approaches, such as summarizing and removing content, could potentially have been more effective for some students; however, this approach carries the risk of oversimplifying the material for those with higher ability levels. A second explanation why text adaptations may not work for students with ID is that linking text information to prior knowledge is essential for recalling content but poses a significant challenge for this group of readers [11], likely a consequence of limited working memory capacity. While lexical and syntactic adaptations may improve decoding and comprehension of individual sentences, adaptations beyond the micro level seem essential to provide the scaffolding needed for comprehension and linking [59]. Examples of macro-level adaptations are strategies to enhance coherence, such as reorganizing text content or employing rhetorical devices like evocations and refutations to activate readers' prior topic knowledge [60]. Several studies on ID have found unexpectedly low comprehension levels, especially for abstract texts requiring expert knowledge [32,35,58]. The number of complicated words that the students reported in this study suggests that the text topics may have been challenging for them. Topic familiarity is crucial for comprehension, especially for struggling readers [61]. The texts used were selected from age-appropriate textbooks, assuming that students would have some prior knowledge. However, we may have overestimated their reading abilities in relation to relatively unknown expository topics.

In iterations 2 and 3, test results across both types of assessments generally indicated improved comprehension for the adapted texts compared to the original versions, with one exception. Although the differences were not statistically significant, we consider the small to moderate effect sizes potentially meaningful, given the limited sample size of only nine participants per iteration. These results correspond to research findings of positive results on adapted texts for students with ID [10,29,30]. Since multiple adaptations were applied simultaneously, it is not possible to determine which specific changes contributed to the improvement. Beginning in iteration 2, the syntactic adaptation tools were enhanced, resulting in more frequent sentence splitting and adaptations to achieve straight word order compared to iteration 1. Additionally, improvements to the synonym replacement corpus led to a slightly higher number of synonym substitutions in some texts. However, a persistent challenge was that many of the words students found difficult were conceptual terms without true synonyms, making them difficult or impossible to replace.

In the third iteration, we kept the adaptations from iteration 2 and focused on discourse-level adaptations and word definitions. While students showed better performance on the adapted texts in the multiple-choice tests, comprehension levels were comparable for both text versions in the SVT test. The discrepancy between the two reading comprehension test results is difficult to interpret. However, the findings suggest that surface-level comprehension may not have been supported by the addition of text-level features such as increased length, bolded keywords, and bullet lists. In addition to being helpful, these features could also interrupt reading flow and thereby comprehension. In general, the results of students taking part in iteration 3 showed higher variability compared to the other student groups, indicated by higher standard deviations.

Evaluations of text and perceived comprehension

In Iteration 1, we found negligible differences in the reading comprehension measures, but student ratings indicated a preference for the adapted texts. Students perceived them as easier, more interesting, and felt they understood them better. In Iteration 2, there were measurable differences in reading comprehension favouring the adapted texts; however, all ratings suggested a preference for the original versions. In Iteration 3, the student performed better on the adapted texts on the multiple-choice questions but rated higher self-assessed comprehension for the original texts along with greater interest in the adapted versions. These patterns are not in line with what we expected, namely higher self-assessed comprehension, lower difficulty ratings, and greater interest for the text versions where students demonstrated better performance. If we focus on the discrepancy between self-assessed reading

comprehension and actual comprehension, the results are likely connected to the impaired metacognitive skills in individuals with ID [13,14]. Meta-comprehension, the awareness of whether one truly understands [62,63], may play a critical role in explaining our results. In terms of magnitude, students typically reported high levels of comprehension and low levels of difficulty. Overestimation of comprehension was also observed in a study by Fajardo et al. [58], where students with ID believed they had answered 80% of the questions correctly, while their actual accuracy was only 55%. It is important to consider what *comprehension* means to the participants themselves. While the ultimate goal of reading is understanding, comprehension is not an all-or-nothing process [62] and often depends on the reader's own criteria. The interviews indicated that students often felt satisfied by learning a new fact or recalling a specific detail from the text. Remembering such a fact may have contributed to a positive self-assessment of their comprehension, even if a deeper understanding was lacking. Furthermore, evaluating the gap between what there is to understand and what one has understood is a complex metacognitive task. The format of the comprehension questions may also contribute to a false sense of success, as it only requires students to select an option or respond with a simple yes or no. Doing so, even without carefully considering the alternatives, can create a sense of completion or satisfaction, despite limited actual understanding.

The preference for the original texts was evident in all three rating questions in iteration 2 and partially in iteration 3. These ratings may reflect students' reactions to the somewhat unfamiliar format of the Easy Language texts. Although few students explicitly expressed this, one noted that the increased length, due to the addition of a summary, made the adapted text more difficult. Other unfamiliar features, such as unusual vocabulary, sentence structures, or formatting, may also have shaped students' perceptions. The adapted texts might have felt less natural, potentially due to their shorter or more fragmented sentence construction.

Thinking and talking about the text

When asked about reading challenges, students mainly mentioned unfamiliar words, particularly complex concepts and compound words. They did not distinguish between decoding difficulties and comprehension issues and did not verbalize any strategies to infer word meanings from context. At the sentence level, they disliked long sentences and the frequent use of commas to separate ideas, likely due to difficulties with working memory. This is similar to challenges seen in individuals with dyslexia, where decoding is not automatic [64]. Positively, they appreciated headings, subheadings, and text formatting, which were the same for both original and adapted text versions. Interestingly, none mentioned sentence structures like subordinate or passive clauses, often flagged as difficult in Easy Language Guidelines [22], perhaps because these features were less noticeable compared to more obvious text elements. Effective remediation of poor comprehension by the reader requires a triggering event, something that stands out as unfamiliar or unexpected [62]. The goal of the metacognitive interviews was to collect triggering events for our target group. When decoding is fluent and understanding is effortless, anomalies are more easily noticed and remembered. However, triggering events are less likely to occur when overall comprehension is low, as was likely the case for this group of students. This may explain why participants identified few issues when discussing the text.

In the third iteration, students expressed positive views on features like summaries and highlighted keywords, but their responses may reflect acquiescence rather than genuine metacognitive insight, as few could explain how these features helped. Acquiescence, a tendency to agree regardless of the question, is common in individuals with ID [65], and may stem from broader patterns of submissiveness and compliance, potentially influencing their feedback.

Effectively using text-level features requires metacognitive abilities and flexibility. While some students attempted to engage with features like revisiting word definitions, they struggled to retain the information due to the challenging concepts in the texts. More critically, they lacked the necessary reading strategies to benefit from these adaptations. While text-level modifications can be helpful, their effectiveness diminishes if readers lack the skills to engage with them fully. Combining text-level adaptations

with interventions that provide explicit instructions on how to use these features may be essential for achieving better outcomes [66,67].

Methods to inform future adaptations

Assessing reading comprehension is critical for evaluating the effectiveness of text adaptations for students with ID. Since their reading abilities are often overestimated [58], ensuring that texts are at an appropriate level and using suitable test formats is essential [68,69]. In this study, the participants scored quite poorly on the multiple-choice items, with comprehension levels ranging from 36% to 59%. We wanted to rule out the possibility that the test format itself was too complex for the students and therefore did not accurately reflect their true comprehension. To address this, we introduced an additional task in Iteration 2. In the Sentence Verification Technique (SVT) task, participants were asked whether they recognized specific sentences from the texts, assessing surface-level understanding compared to the inferential demands of the multiple-choice questions. Thus, this task simplified both the question complexity and the response format. With this format, students achieved between 63% and 79% correct answers, suggesting that the SVT task may better align with their actual comprehension levels. However, it is important to note that both assessment formats are likely influenced by guessing. In the SVT task in particular, participants have a 50% chance of selecting the correct answer by guessing alone.

The findings from the reading comprehension tests guided further adaptations at the word and sentence levels in iteration 2. In iteration 3, interview data and limited possibilities to further develop the synonym function prompted a shift in focus. In retrospect, discussing the thoughts behind the selection of an answer with the students could have been informative. As we have discussed previously, evoking metacognitive reasoning is difficult but necessary to understand the thought behind a selection, and what it says about comprehension and strategies.

Even though post-reading perception of text ratings may seem ineffective as an outcome measure, as the participants overestimated their abilities due to weak meta-comprehension, this was important information for us in trying to understand comprehension and metacognition of the participants in combination with the other methods. However, these results did not affect later adaptations.

The most valuable insights came from the discussions, which revealed difficult words and guided our word substitution strategy. Interviews highlighted how memory limitations affected comprehension, but participants did not express clear preferences about sentence structure, so we relied on Easy Language guidelines [22]. Although the students expressed that the features were positive, we saw limited engagement with discourse-level and word explanation adaptations applied in iteration 3.

In summary, each assessment presented its own challenges, yet all contributed to a deeper understanding of the students' comprehension. This aligns with the recommendation to use triangulation when working with this population [70]. Although we gained some insights into improving text adaptations, applying these findings to make the texts easier proved difficult, likely in part due to the initial overestimation of the students' reading comprehension when selecting texts.

Limitations

In this study, we conducted extensive testing of text comprehension but did not include standardized measures of reading comprehension or IQ due to test prioritization. This represents a limitation as it may reduce the generalizability of the findings. While IQ levels were in the mild to moderate range, a more precise assessment would likely have provided a valuable context for interpreting individual differences in reading comprehension. Similarly, the absence of a reading comprehension baseline limits our ability to determine whether the effects of the adaptations were influenced by reading ability, as suggested in previous research.

We examined the effectiveness of different adaptation strategies; however, each iteration involved a new group of participants, making direct comparisons less reliable. Although the groups were broadly similar in cognitive, linguistic, and age profiles, one group consisted almost entirely of male participants, which may have influenced the results.

Conclusions

This exploratory study aimed to evaluate how students with ID read and understood ATA texts compared to the original versions. We used three rounds of testing on new groups of students and improved text adaptations based on results from previous rounds. We found that syntactic and lexical adaptations can make a difference in comprehension results, but the effect of adding discourse-level adaptations and giving word definitions was less clear. The students' ratings of perceived difficulty, self-assessed reading comprehension, and interest did not align with their actual reading comprehension results. This discrepancy may be explained by limited metacognitive skills and a preference for texts that resemble more typical or familiar formats. While the students mentioned having trouble with difficult concepts and long sentences, their limited ability to reflect on their understanding made it hard for them to point out exactly what needed to be changed. Still, talking to students about what they found difficult was very helpful for guiding further improvements. Overall, the findings suggest that ATA can be helpful, but students likely need additional support when reading factual or informational texts.

Acknowledgements

The authors would like to thank Cecilia Ivarsson and Maximilian Horn for all their help with the data collection.

Disclosure statement

The authors declare no conflicts of interest related to this work.

Funding

This study was funded by Vetenskapsrådet [the Swedish research council], Grant/Award Number 2020-00151.

References

- [1] Arvidsson J. Sysselsättning och social rättvisa: en nationell registerstudie om 12 269 unga vuxna med intellektuell funktionsnedsättning [Employment and social justice: a national registry study of 12,269 young adults with intellectual disabilities]. Umeå Universitet. 2016.
- [2] Luthra R, Westberg N, Högdin S, et al. 'Just because one has attended a special class does not mean that one isn't capable': the experiences of becoming and being not involved in traditional occupations for young people with intellectual disability. *Nordic Social Work Res.* 2020;10(4):330–342. doi: [10.1080/2156857X.2018.1553734](https://doi.org/10.1080/2156857X.2018.1553734).
- [3] American Psychiatric Association & DSM-5 Task Force. *Diagnostic and statistical manual of mental disorders: DSM-5*. 5th ed. Washington, (DC). American Psychiatric Publishing; 2013. doi: [10.1176/appi.books.9780890425596](https://doi.org/10.1176/appi.books.9780890425596).
- [4] Danielsson H, Henry L, Messer D, et al. Strengths and weaknesses in executive functioning in children with intellectual disability. *Res Dev Disabil.* 2012;33(2):600–607. doi: [10.1016/j.ridd.2011.11.004](https://doi.org/10.1016/j.ridd.2011.11.004).
- [5] Nilsson K, Danielsson H, Elwér Å, et al. Investigating reading comprehension in adolescents with intellectual disabilities: evaluating the simple view of reading. *J Cogn.* 2021;4(1):56. doi: [10.5334/joc.188](https://doi.org/10.5334/joc.188).
- [6] Van der Molen MJ, Henry LA, Van Luit JEH. Working memory development in children with mild to borderline intellectual disabilities. *J Intellect Disabil Res.* 2014;58(7):637–650. doi: [10.1111/jir.12061](https://doi.org/10.1111/jir.12061).
- [7] Lemons CJ, Zigmond N, Kloo AM, et al. Performance of students with significant cognitive disabilities on early-grade curriculum-based measures of word and passage reading fluency. *Except Child.* 2013;79(4):408–426. doi: [10.1177/001440291307900402](https://doi.org/10.1177/001440291307900402).
- [8] Wei X, Blackorby J, Schiller E. Growth in reading achievement of students with disabilities ages 7 to 17. *Exceptional Children.* 2011;78(1):89–106. doi: [10.1177/001440291107800106](https://doi.org/10.1177/001440291107800106).
- [9] Nilsson K, Elwér Å, Messer D, et al. Developmental trajectories of reading ability in adolescents with intellectual disabilities. *Int J Disabil Develop Educ.* 2024;1–17. doi: [10.1080/1034912X.2024.2403391](https://doi.org/10.1080/1034912X.2024.2403391).
- [10] Fajardo I, Ávila V, Ferrer A, et al. Easy-to-read texts for students with intellectual disability: linguistic factors affecting comprehension. *J Appl Res Intellect Disabil.* 2014;27(3):212–225. doi: [10.1111/jar.12065](https://doi.org/10.1111/jar.12065).
- [11] van Wingerden E, Segers E, van Balkom H, et al. Cognitive and linguistic predictors of reading comprehension in children with intellectual disabilities. *Res Dev Disabil.* 2014;35(11):3139–3147. doi: [10.1016/j.ridd.2014.07.054](https://doi.org/10.1016/j.ridd.2014.07.054).
- [12] Kintsch W. The role of knowledge in discourse comprehension: a construction-integration model. *Psychol Rev.* 1988;95(2):163–182. doi: [10.1037/0033-295X.95.2.163](https://doi.org/10.1037/0033-295X.95.2.163)

- [13] Igier S, Pennequin V. Roles of the metacognition and emotional systems in a categorization task for adults with moderate and severe learning disabilities. *Int J Develop Disab*. 2022;68(3):281–289. doi: [10.1080/20473869.2020.1759952](https://doi.org/10.1080/20473869.2020.1759952).
- [14] Pennequin V, Igier S, Pivry S, et al. Metacognition and emotion in adolescents with intellectual disability: links with categorization performance. *Curr Psychol*. 2023;42(2):1013–1021. doi: [10.1007/s12144-021-01493-1](https://doi.org/10.1007/s12144-021-01493-1).
- [15] Alfassi M, Weiss I, Lifshitz H. The efficacy of reciprocal teaching in fostering the reading literacy of students with intellectual disabilities. *European J Special Needs Edu*. 2009;24(3):291–305. doi: [10.1080/08856250903016854](https://doi.org/10.1080/08856250903016854).
- [16] Allor JH, Mathes PG, Roberts JK, et al. Is scientifically based reading instruction effective for students with below-average IQs? *Except Child*. 2014;80(3):287–306. doi: [10.1177/0014402914522208](https://doi.org/10.1177/0014402914522208).
- [17] Dessemontet RS, Martinet C, Chambrier A-F, et al. A meta-analysis on the effectiveness of phonics instruction for teaching decoding skills to students with intellectual disability. *Edu Res Rev*. 2019;26:52–70. doi: [10.1016/j.edurev.2019.01.001](https://doi.org/10.1016/j.edurev.2019.01.001).
- [18] van den Bos KP, Nakken H, Nicolay PG, et al. Adults with mild intellectual disabilities: can their reading comprehension ability be improved? *J Intellect Disabil Res*. 2007;51(Pt 11):835–849. doi: [10.1111/j.1365-2788.2006.00921.x](https://doi.org/10.1111/j.1365-2788.2006.00921.x).
- [19] Crossley SA, Louwerse MM, McCarthy PM, et al. A linguistic analysis of simplified and authentic texts. *Modern Lang J*. 2007;91(1):15–30. doi: [10.1111/j.1540-4781.2007.00507.x](https://doi.org/10.1111/j.1540-4781.2007.00507.x).
- [20] Inclusion Europe. Easy-to-read checklist: checklist to make sure your document is easy to read. 3rd ed. Brussels, Belgium: Inclusion Europe; 2020.
- [21] Nomura M, Skat Nielsen G, Trönbacke B. Guidelines for easy-to-read materials. *Int Fed Lib Assoc Inst (IFLA)*. 2010.
- [22] PLAIN. Federal plain language guidelines (Rev. 1); 2011. Available from: <https://www.plainlanguage.gov/media/FederalPLGuidelines.pdf>.
- [23] Swedish Agency for Accessible Media [Myndigheten för tillgängliga medier]. How to write easy-to-read public information. *Så Skriver Du Lättläst Samhällsinformation*; 2023. Available from: <https://www.mtm.se/produkter-och-tjanster/produkter/sprakguider/sa-skriver-du-lattlast-samhallsinformation/>.
- [24] Wengelin Å. Mot en evidensbaserad språkvård? En kritisk granskning av några svenska klarspråksråd i ljuset av forskning om läsbarhet och språkbearbetning [Towards evidence-based language advice? A critical review of some Swedish plain language guidelines in light of research on readability and language processing]. *Sakprosa*. 2015;7(2):17. doi: [10.5617/sakprosa.983](https://doi.org/10.5617/sakprosa.983).
- [25] Rennes E. Automatic adaptation of Swedish text for increased inclusion. Linköping: Linköping University Electronic Press; 2022. doi: [10.3384/9789179292706](https://doi.org/10.3384/9789179292706).
- [26] Papineni K, Roukos S, Ward T, et al. BLEU: a method for automatic evaluation of machine translation. *Proceedings of the 40th Annual Meeting on Association for Computational Linguistics*; 2002, p. 311–318.
- [27] Xu W, Napoles C, Pavlick E, et al. Optimizing statistical machine translation for text simplification. *TACL*. 2016;4:401–415. doi: [10.1162/tacl_a_00107](https://doi.org/10.1162/tacl_a_00107).
- [28] Sulem E, Abend O, Rappoport A. Semantic structural evaluation for text simplification. *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1*; 2018. p. 685–696. doi: [10.18653/v1/N18-1063](https://doi.org/10.18653/v1/N18-1063).
- [29] Karreman J, Van Der Geest T, Buursink E. Accessible website content guidelines for users with intellectual disabilities. *Res Intel Disabil*. 2007;20(6):510–518. doi: [10.1111/j.1468-3148.2006.00353.x](https://doi.org/10.1111/j.1468-3148.2006.00353.x).
- [30] Säuberli A, Holzknacht F, Haller P, et al. Digital comprehensibility assessment of simplified texts among persons with intellectual disabilities. *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*; 2024. doi: [10.1145/3613904.3642570](https://doi.org/10.1145/3613904.3642570).
- [31] Fajardo I, Tavares G, Ávila V, et al. Towards text simplification for poor readers with intellectual disability: when do connectives enhance text cohesion? *Res Dev Disabil*. 2013;34(4):1267–1279. doi: [10.1016/j.ridd.2013.01.006](https://doi.org/10.1016/j.ridd.2013.01.006).
- [32] Poncelas A, Murphy G. Accessible information for people with intellectual disabilities: Do symbols really help? *Res Intel Disabil*. 2007;20(5):466–474. doi: [10.1111/j.1468-3148.2006.00334.x](https://doi.org/10.1111/j.1468-3148.2006.00334.x).
- [33] Hurtado B, Jones L, Burniston F. Is easy read information really easier to read? *J Intellect Disabil Res*. 2014;58(9):822–829. doi: [10.1111/jir.12097](https://doi.org/10.1111/jir.12097).
- [34] Falk L, Johansson S. Hur fungerar lättlästa texter på webben?. [How do easy-to-read texts work on the web?]. Funka; 2006. Available from: https://www.funka.com/contentassets/755a917870714fc8b9dd5a34de6a3237/rapport_lattlast.pdf.
- [35] Frondén C, Kaakinen J. Reading easy language texts written by public authorities: evidence from eye tracking. *FJL*. 2023;36: 7–36. doi: [10.61197/fjl.126062](https://doi.org/10.61197/fjl.126062).
- [36] Dunn LM, Dunn M. Peabody picture vocabulary test III. Circle Pines (MN): American Guidance Service; 1997.
- [37] Elwér Å, Fridolfsson I, Samuelsson S, et al. LÄST – test i läsförståelse, läsning och stavning för årskurs 1–6 [LÄST – a test of reading comprehension, reading, and spelling for grades 1–6]. Stockholm: Hogrefe; 2016.
- [38] Bishop D. Test for the reception of grammar (TROG). Cambridge (UK): Medical Research Council; 1989.
- [39] Wechsler D. WISC-V: Administration and Scoring Manual. San Antonio (TX): PsychCorp. 2014.
- [40] Flising LL. Hem-och konsumentkunskap 7–9 [Home and consumer studies, grades 7–9]. Stockholm: Liber; 2018.
- [41] Mårtensson S, Lindberg L. Geografi Ämnesboken [Geography Subject Book]. 3rd ed. Stockholm: Liber; 2012.
- [42] Heimann Muhlenbock K. I see what you mean [University of Gothenburg]; 2013. Available from: <https://gupea.ub.gu.se/handle/2077/32472>.

- [43] Kann V, Rosell M. Free construction of a free Swedish dictionary of synonyms. Proceedings of the 15th NODALIDA Conference; 2005. p. 105–110.
- [44] Keskisärkkä R, Jönsson A. Investigations of synonym replacement for Swedish. Northern Euro J Lang Technol. 2013;3(3):41–59.
- [45] Johansson V, Rennes E. Automatic extraction of synonyms from an easy-to-read corpus. Proceedings of the Sixth Swedish Language Technology Conference (SLTC-16), Umeå, Sweden; 2016.
- [46] Rennes E, Jönsson A. A tool for automatic simplification of swedish texts. Proceedings of the 20th Nordic Conference of Computational Linguistics (NoDaLiDa-2015), Vilnius, Lithuania; 2015.
- [47] Royer JM, Hastings CN, Hook C. A sentence verification technique for measuring reading comprehension. J Read Behav. 1979;11(4):355–363. doi: [10.1080/10862967909547341](https://doi.org/10.1080/10862967909547341).
- [48] Royer JM, Carlo MS. The assessment of levels of domain expertise while reading. J Educ Psychol. 1996;14(3):171–179. doi: [10.1207/s1532690xci1403](https://doi.org/10.1207/s1532690xci1403).
- [49] Holmer D, Rennes E. NyLLex: a novel resource of swedish words annotated with reading proficiency level. Proceedings of the Language Resources and Evaluation Conference; 2022. p. 1326–1331. <http://www.lrec-conf.org/proceedings/lrec2022/pdf/2022.lrec-1.141.pdf>
- [50] Bojanowski P, Grave E, Joulin A, et al. Enriching word vectors with subword information. TACL. 2017;5:135–146. doi: [10.1162/tac1_a_00051](https://doi.org/10.1162/tac1_a_00051).
- [51] Holmer D, Jönsson A. 3rd Workshop on Tools and Resources for People with REAding Difficulties (READI 2024). Proceedings of the Auxiliary Techniques to Help Readers Understand Texts; 2024. <https://aclanthology.org/2024.readi-1.pdf>
- [52] Priscepov R. Automatic text simplification of Swedish for second language learners [Master's thesis]. Technische Universität München; 2020.
- [53] Grootendorst M. 2021. MaartenGr/KeyBERT: bibTeX (Version v0.1.3) [Computer software]. Zenodo. doi: [10.5281/zenodo.4461265](https://doi.org/10.5281/zenodo.4461265).
- [54] Ekgren A, Gyllensten AC, Stollenwerk F, et al. GPT-SW3: an autoregressive language model for the Scandinavian languages. Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024); 2024. p. 7886–7900. <https://urn.kb.se/resolve?urn=urn:nbn:se:kt:h:diva-354954>
- [55] Andersson E, Jönsson A. Methods for increasing cohesion in automatically extracted summaries of Swedish news articles. Proceedings of the 13th International Conference on Language Resources and Evaluation (LREC 2022); 2022.
- [56] Smith C, Jönsson A. Automatic summarization as means of simplifying texts, an evaluation for Swedish. 18th Nordic Conference of Computational Linguistics; 2011.
- [57] Decker A. Towards automatic grammatical simplification of Swedish text [Master's thesis]. Stockholm University; 2003.
- [58] Fajardo I, Ávila V, Delgado P, et al. Video-blogs and linguistic simplification for students with intellectual disability. J Appl Res Intellect Disabil. 2022;35(5):1217–1230. doi: [10.1111/jar.13016](https://doi.org/10.1111/jar.13016).
- [59] Maaß C, Rink I. Scenarios for easy language translation: how to produce accessible content for users with diverse needs. In S Hansen-Schirra, C Maaß, editors. Easy language research: text and user perspectives. Berlin: Frank & Timme; 2020. p. 123–145.
- [60] Arfé B, Mason L, Fajardo I. Simplifying informational text structure for struggling readers. Read Writ. 2018;31(9):2191–2210. doi: [10.1007/s11145-017-9785-6](https://doi.org/10.1007/s11145-017-9785-6).
- [61] Smith R, Snow P, Serry T, et al. The role of background knowledge in reading comprehension: a critical review. Reading Psychol. 2021;42(3):214–240. doi: [10.1080/02702711.2021.1888348](https://doi.org/10.1080/02702711.2021.1888348).
- [62] Brown AL. Metacognitive development and reading. In RJ Spiro, BB Bruce, WF Brewer, editors. Theoretical issues in reading comprehension. Hillsdale (NJ): Lawrence Erlbaum; 1980. p. 453–481.
- [63] Flavell JH. Metacognitive aspects of problem solving. In LB Resnick, editor. The nature of intelligence. Hillsdale (NJ): Lawrence Erlbaum; 1976. p. 231–235.
- [64] Simmons F, Singleton C. The reading comprehension abilities of dyslexic students in higher education. Dyslexia. 2000;6(3):178–192. doi: [10.1002/1099-0909\(200007/09\)6:3<178::AID-DYS171>3.0.CO;2-9](https://doi.org/10.1002/1099-0909(200007/09)6:3<178::AID-DYS171>3.0.CO;2-9).
- [65] Finlay WML, Lyons E. Acquiescence in interviews with people who have mental retardation. Mental Retardation. 2002;40(1):14–29. doi: [10.1352/0047-6765\(2002\)040<0014:AIWPW>2.0.CO;2](https://doi.org/10.1352/0047-6765(2002)040<0014:AIWPW>2.0.CO;2).
- [66] Buell S, Langdon PE, Pounds G, et al. An open randomized controlled trial of the effects of linguistic simplification and mediation on the comprehension of 'Easy Read' text by people with intellectual disabilities. J Appl Res Intellect Disabil. 2020;33(2):219–231. doi: [10.1111/jar.12666](https://doi.org/10.1111/jar.12666).
- [67] Ladas G, Iatraki G, Souli SG. Providing access to reading comprehension for greek secondary students with mild intellectual disabilities. Educ Sci. 2022;12(12):921. doi: [10.3390/educsci12120921](https://doi.org/10.3390/educsci12120921).
- [68] Finlay WML, Lyons E. Methodological issues in interviewing and using self-report questionnaires with people with mental retardation. Psychol Assess. 2001;13(3):319–335. doi: [10.1037/1040-3590.13.3.319](https://doi.org/10.1037/1040-3590.13.3.319).
- [69] Hollomotz A. Successful interviews with people with intellectual disability. Qualit Res. 2018;18(2):153–170. doi: [10.1177/1468794117713810](https://doi.org/10.1177/1468794117713810).
- [70] Llewellyn G. Qualitative research with people with intellectual disability. Occupat Therapy Intl. 1995;2(2):108–127. doi: [10.1002/oti.6150020206](https://doi.org/10.1002/oti.6150020206).